PART 5
Electromagnetic Character of Light

This part consists of three chapters discussing various aspects of the electromagnetic character of light waves. In the first chapter of this part (Chapter 22) the generation and analysis of various forms of polarized light is discussed followed by a detailed analysis of propagation of electromagnetic waves in anisotropic media including first principle derivations of wave and ray velocities. Applications like optical activity, Faraday rotation etc., have also been discussed. Chapter 23 is a bit mathematical – starting with Maxwell's equations, various states of polarization have been discussed; the wave equation has also been derived which had led Maxwell to predict the existence of electromagnetic waves. Reflection and refraction of electromagnetic waves by a dielectric interface has been discussed in Chapter 24. The results directly explain phenomena like Brewster's law, total internal reflection, evanescent waves, Fabry–Perot transmission resonances, etc.

Chapter Twenty Two

POLARIZATION AND DOUBLE REFRACTION

... As to the other emanation which should produce the irregular refraction, I wished to try what Elliptical waves, or rather spheroidal waves, would do; and these I suppose would spread differently both in the ethereal matter diffused throughout the crystal and in the particles of which it is composed...

Christiaan Huygens

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: know how to produce various forms of polarized light waves.
LO 2: discuss superposition of two disturbances and their mathematical analysis.
LO 3: describe the phenomenon of double refraction.
LO 4: know about the interference of polarized light.
LO 5: analyze polarized light.
LO 6: discuss optical activity.
LO 7: discuss change in state of polarization (SOP) of light beam propagating through an elliptic core single-mode optical fiber.
LO 8: discuss the working of Wollaston and Rochon prisms for producing linearly polarized waves.
LO 9: analyze propagation by electromagnetic waves in anisotropic media.
LO 10: explain ray velocity and ray refractive index.
LO 11: use Jones calculus to study the propagation of polarized waves through various plates.
LO 12: discuss Faraday rotation and its applications.
LO 13: understand the theory of optical activity.
LO 14: understand theory of Faraday rotation.

Important Milestones

1669 *Erasmus Bartholinus discovered double refraction in calcite.*

1678 *In the wave theory of light communicated to the Academie des Science in Paris, Christiaan Huygens gave the theory of double refraction in calcite discovered by Bartholinus.*

1815 *David Brewster showed polarization of light by reflection.*

1828 *William Nicol invented the prism which produced polarized light - this prism came to be known as the Nicol Prism.*

1929 *Edwin Land, an American scientist and inventor, patented Polaroid which is the name of a type of synthetic plastic sheet which is used to polarize light.*

22.1 INTRODUCTION LO 1

If we oscillate one end of a string up and down then a transverse wave is generated [see Fig. 22.1(a)]. Each point of the string executes a sinusoidal oscillation in a straight line (along the x-axis) and the wave is, therefore, known as a **linearly polarized wave**. It is also known as a **plane polarized wave** because the string is always confined to the x-z plane. The displacement for such a wave can be written in the form

$$\left.\begin{aligned} & x(z,t) = a\cos(kz-\omega t+\phi_1) \\ \text{and} \quad & y(z,t) = 0 \end{aligned}\right\} \tag{22.1}$$

where a represents the amplitude of the wave and ϕ_1 is the phase constant to be determined from the instant we choose as $t=0$; the y-coordinate of the displacement is always zero. At any instant the displacement will be a cosine curve as shown in Fig. 22.1(a). Further, an arbitrary point $z=z_0$ will execute a simple harmonic motion of amplitude a. The string can also be made to vibrate in the y-z plane [see Fig. 22.1(b)] for which the displacement would be given by

$$\left.\begin{aligned} & x(z,t) = 0 \\ \text{and} \quad & y(z,t) = a\cos(kz-\omega t+\phi_2) \end{aligned}\right\} \tag{22.2}$$

In general, the string can be made to vibrate in any plane containing the z-axis. If one rotates the end of the string on the circumference of a circle then each point of the string will move in a circular path as shown in Fig. 22.2; such a wave is known as a circularly polarized wave and the corresponding displacement would be given by

$$\left.\begin{aligned} & x(z,t) = a\cos(kz-\omega t+\phi) \\ \text{and} \quad & y(z,t) = -a\sin(kz-\omega t+\phi) \end{aligned}\right\} \tag{22.3}$$

so that $x^2 + y^2$ is a constant ($= a^2$). As we will see later, Eq. (22.3) represents a right circularly polarized wave.

We next consider a long narrow slit placed in the path of the string as shown in Fig. 22.3(a). If the length of the slit is along the direction of the displacement then the entire amplitude will be transmitted as shown in Fig. 22.3(a). On the other hand, if the slit is at right angle to the direction of the displacement, then almost nothing will be transmitted to the other side of the slit [see Fig. 22.3(b)]. This is because of the fact that the slit allows only the component of the displacement, which is along the length of the slit, to pass through; as such, if a longitudinal wave was propagating through the string then the amplitude of the transmitted wave would have been the same for all orientations of the slit. Thus, the change in the amplitude of the transmitted wave with the orientation of the slit is due to the transverse character of the wave. Indeed, an experiment which is, in principle, very similar to the experiment discussed above proves the transverse character of light waves. However, before we discuss the experiment with light waves we must define an unpolarized wave.

We once again consider transverse waves generated at one end of a string. If the plane of vibration is changed in a random manner in very short intervals of time, then such a wave is known as an unpolarized wave*. If an unpolarized wave falls on a slit S_1 (see Fig. 22.4) then the displacement associated with the transmitted wave will be along the length of the slit and a rotation of the slit will not affect the amplitude of the transmitted wave although the plane of

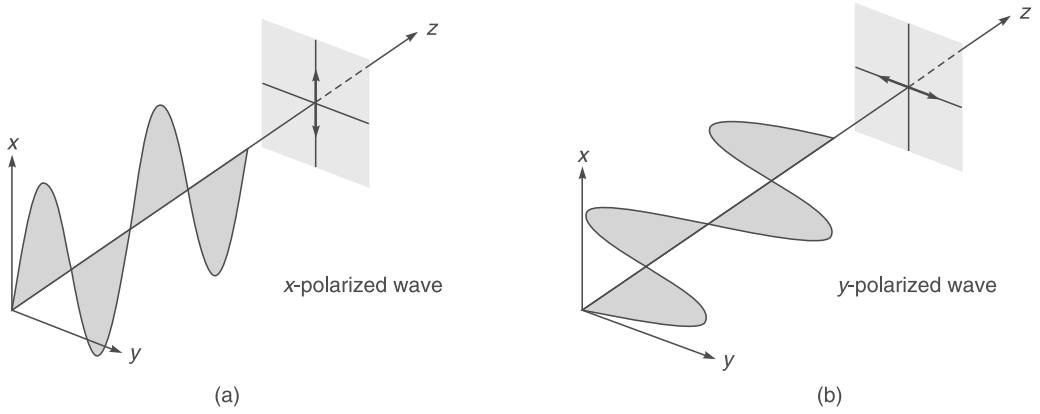


Fig. 22.1 (a) A linearly polarized wave on a string with the displacement confined to the x-z plane; (b) A linearly polarized wave on a string with the displacement confined to the y-z plane.

* By a short interval, we imply times which are short compared to the detection time, however, for the wave to be characterized with a certain frequency ν, this time has to be much greater than $1/\nu$, so that in the short interval it executes a large number of oscillations (see also Sec. 17.1).

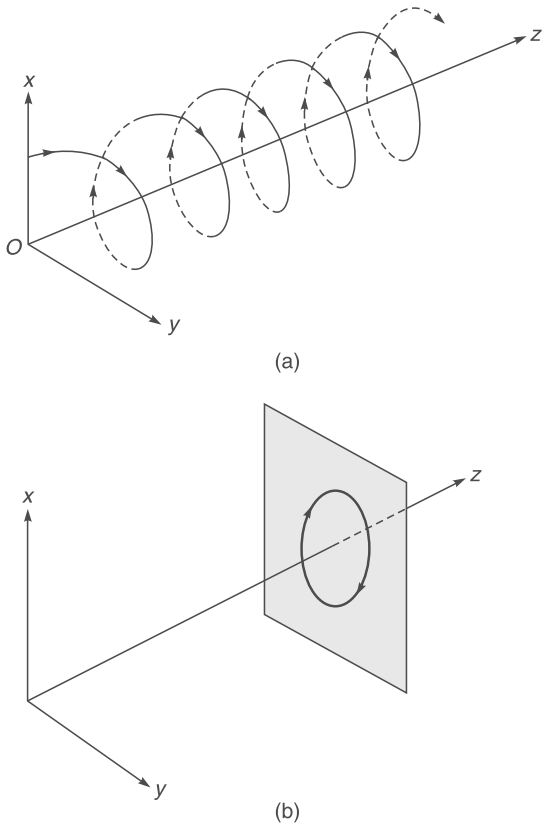


Fig. 22.2 (a) The displacement corresponding to a circularly polarized wave – all points on the string are at the same distance from the z-axis. (b) Each point on the string rotates on the circumference of the circle.

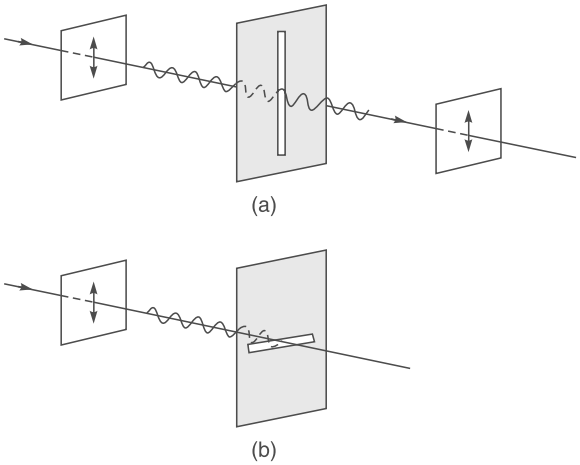


Fig. 22.3 If a linearly polarized transverse wave (propagating on a string) is incident on a long narrow slit, then the slit will allow only the component of the displacement, which is along the length of the slit, to pass through.

polarization of transmitted wave depend on the orientation of the slit (see Fig. 22.4). Thus, the transmitted wave will be linearly polarized and the slit S_1 is said to act as a polarizer. If this polarized beam falls on another slit S_2 (see Fig. 22.4), then by rotating the slit S_2, we obtain a variation of the transmitted amplitude as discussed earlier; the second slit is said to act as an analyzer.

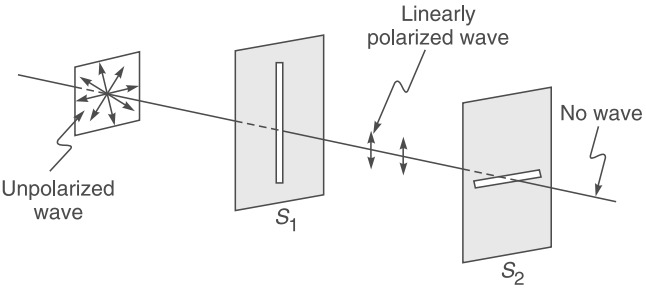


Fig. 22.4 If an unpolarized wave propagating on a string is incident on a long narrow slit S_1, then the transmitted beam will be linearly polarized and its amplitude will not depend on the orientation of S_1. If this polarized wave is allowed to pass through another slit S_2, then the intensity of the emerging wave will depend on the relative orientation of S_2 with respect to S_1.

The transverse character of light waves was known in the early years of the nineteenth century; however, the nature of the displacement associated with a light wave was known only after Maxwell had put forward his famous electromagnetic theory. We will discuss the basic electromagnetic theory in the next chapter where we will show that associated with a plane electromagnetic wave there is an electric field **E** and a magnetic field **H** which are at right angles to each other. For a linearly polarized wave propagating in the z direction (in a dielectric) the electric and magnetic fields can be written in the form [see Fig. 22.5]

$$E_x = E_0 \cos(kz - \omega t),\ E_y = 0,\ \ E_z = 0 \qquad \textbf{(22.4)}$$

and $$H_x = 0,\ H_y = H_0 \cos(kz - \omega t),\ \ H_z = 0 \qquad \textbf{(22.5)}$$

where

$$k = \frac{\omega}{v} = \omega \sqrt{\varepsilon \mu} \qquad \textbf{(22.6)}$$

and

$$v = \frac{1}{\sqrt{\varepsilon \mu}} \qquad \textbf{(22.7)}$$

represents the velocity of the waves, ε and μ are the dielectric permittivity and the magnetic permeability of the medium. Since $E_z = 0$ and $H_z = 0$, the wave is transverse. Equations (22.4) and (22.5) also show that **E** and **H** are at

right angles to each other and both the vectors are at right angles to the direction of propagation (which is along the z-axis). In fact, the direction of propagation is along the vector ($\mathbf{E} \times \mathbf{H}$). Electromagnetic theory also tells us that for a dielectric [see Sec. 23.3]:

$$H_0 = \frac{k}{\omega\mu} E_0 = \frac{1}{\eta_0 / n} E_0 \qquad \textbf{(22.8)}$$

where

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = c\mu_0 \approx 120\pi \quad \text{ohms}$$

is the intrinsic impedance of free space and n is the refractive index of the dielectric (see Sec. 23.3).

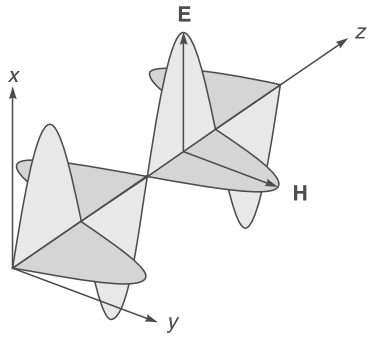


Fig. 22.5 An x-polarized electromagnetic wave propagating in the z-direction.

We consider an ordinary light beam falling on a Polaroid P_1 as shown in Fig. 22.6; a Polaroid is a plastic-like material used for producing polarized light—it will be discussed in detail in the next section. In general, an ordinary light beam (like the one coming from a sodium lamp or from the sun) is unpolarized, i.e., the electric vector (in a plane transverse to the direction of propagation) keeps changing its direction in a random manner (see Fig. 22.6). When such a beam is incident on a Polaroid the emergent light is linearly polarized with its electric vector oscillating in a particular direction as shown in Fig. 22.6. The direction of the electric vector of the emergent beam will depend on the orientation of the Polaroid. As will be shown in Sec. 22.3.1 the component of **E** along a particular direction gets absorbed by the Polaroid and the component at right angles to it passes through. The direction of the electric vector of the emergent wave is usually called the pass axis of the Polaroid. If the Polaroid P_2 is absent and if the Polaroid P_1 is rotated about the z-axis, there will be no variation of intensity. However, if we place another Polaroid P_2, then by rotating the Polaroid P_2 (about the z-axis) one will observe variation of intensity and at two positions there

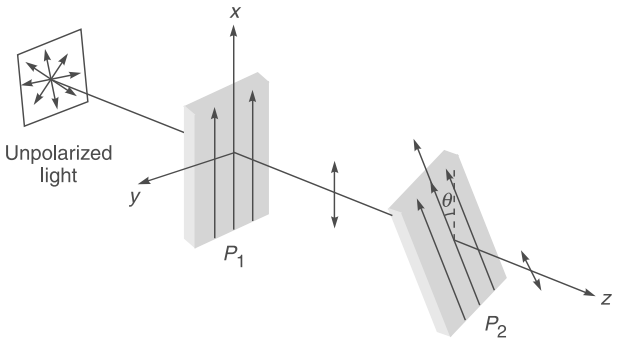


Fig. 22.6 For an unpolarized wave propagating in the $+z$-direction, the electric vector (which lies in the x-y plane) continues to change its direction in a random manner. If an unpolarized light beam is allowed to fall on a Polaroid, then the emerging beam will be linearly polarized; i.e., the electric vector will oscillate along a particular direction. If we place another Polaroid P_2, then the intensity of the transmitted light will depend on the relative orientation of P_2 with respect to P_1.; if the pass axis of the second polaroid P_2 makes an angle θ with the x-axis, the intensity of the emerging beam will vary as $\cos^2\theta$.

will be almost complete darkness (see Fig. 22.7). A similar phenomenon will also be observed if instead of rotating the Polaroid P_2 we rotate P_1. On the basis of our earlier discussions, this phenomenon proves the transverse character of light; i.e., the displacement associated with a light wave is at right angles to the direction of propagation



Fig. 22.7 Actual photographs with two Polaroids at different angles of relative orientation. (a) If the two Polaroids are parallel to each other, almost the entire light passes through. (b) When the two Polaroids are oriented at with respect to each other about 50% of the light passes through grey. (c) When the two Polaroids are at right angles to each other (notice the position of the grey dot) almost no light will pass through. Photographs adapted from www.a-levelphysicstutor.com; used with kind permission from Dr. Alan J. Reed. See Fig. 26 in the prelim pages.

of the wave. The Polaroid P_1 acts as a polarizer and the transmitted beam is linearly polarized. The second Polaroid acts as an analyzer.

22.2 MALUS' LAW LO 1

Let us consider a Polaroid P_1 which has a pass-axis parallel to the x-axis (see Fig. 22.6); i.e., if an unpolarized beam propagating in the z direction is incident on the Polaroid, then the electric vector associated with the emergent wave will oscillate along the x-axis. We next consider the incidence of the x-polarized beam on the Polaroid P_2 whose pass axis makes an angle θ with the x-axis (see Fig. 22.6). If the amplitude of the incident electric field is E_0, then the amplitude of the wave emerging from the polaroid P_2 will be $E_0 \cos\theta$ and thus the intensity of the emerging beam will be given by

$$I = I_0 \cos^2\theta \qquad \textbf{(22.9)}$$

where I_0 represents the intensity of the emergent beam when the pass axis of P_2 is also along the x-axis (i.e., when $\theta = 0$). Equation (22.9) represents Malus' Law. Thus, if a linearly polarized beam is incident on a Polaroid and if the Polaroid is rotated about the z-axis, then the intensity of the emergent wave will vary according to the above law. For example, if the Polaroid P_2 shown in Fig. 22.6 is rotated in the clockwise direction, then the intensity will increase till the pass-axis is parallel to the x-axis; a further rotation will result in a decrease in intensity till the pass-axis is parallel to the y-axis, where the intensity will be almost zero. If we further rotate it, it will pass through a maximum and again a minimum before it reaches its original position.

Figure 22.7 shows actual photographs of two Polaroids at different relative orientations. In Fig. 22.7 (a) the two are parallel to each other and therefore almost the entire light passes through. In Fig. 22.7 (b) the two Polaroids are oriented at 45° with respect to each other and about 50% of the light passes through; because according to Malus' law $I = I_0 \cos^2 45 = \frac{1}{2} I_0$. Finally, in In Fig. 22.7 (c) the two Polaroids are at right angles to each other (notice the position of the blue dot) and almost no light passes through because $I = I_0 \cos^2 90^\circ = 0$ [see also Fig. 26 in the prelim pages of the book]

22.3 PRODUCTION OF POLARIZED LIGHT LO 1

In this section we will discuss various methods for producing linearly polarized light waves.

22.3.1 The Wire Grid Polarizer and the Polaroid

The physics behind the working of the wire grid polarizer is probably the easiest to understand. It essentially consists of a large number of thin copper wires placed parallel to each other as shown in Fig. 22.8. When an unpolarized electromagnetic wave is incident on it, then the component of the electric vector along the length of the wire is absorbed. This is due to the fact that the electric field does work on the electrons inside the thin wires and the energy associated with the electric field is lost in the Joule heating of the wires. On the other hand, (since the wires are assumed to be very thin) the component of the electric vector along the x-axis passes through without much attenuation. Thus, the emergent wave is linearly polarized with the electric vector along the x-axis. However, for the system to be effective (i.e., for the E_y component to be almost completely attenuated) the spacing between the wires should be less than λ. Clearly, the fabrication of such a polarizer for a 3 cm microwave is relatively easy because the spacing has to be less than 3 cm. On the other hand, since the light waves are associated with a very small wavelength $(\sim 0.5\ \mu\text{m})$, the fabrication of a polarizer in which the wires are placed at distances less than $0.5\ \mu\text{m}$ is extremely difficult. Nevertheless, Bird and Parrish did succeed in putting about 30,000 wires in about one inch; for further details see Refs. 22.1 and 22.2. The details of the procedure for making this wire grating are also discussed in Ref. 22.1. The original work of Bird and Parrish was published in 1950 (see Ref. 22.2).

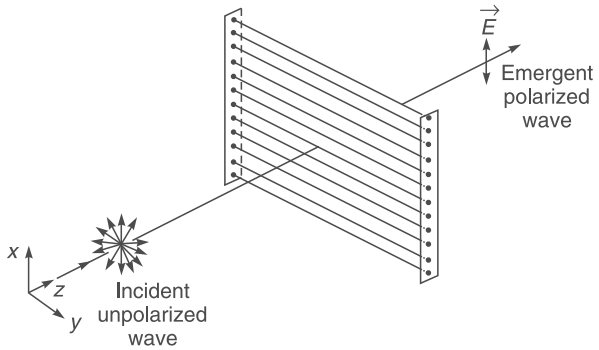


Fig. 22.8 The wire-grid polarizer.

As already pointed out, it is extremely difficult to fabricate a wire grid polarizer which would be effective for visible light. However, instead of long thin wires, one may employ long chain polymer molecules that contain atoms (like iodine) which provide high conductivity along the length of the chain. These long chain molecules are aligned so that they are almost parallel to each other. Because of the high conductivity provided by the iodine atoms, the electric field